Tadelola Aduloju Burgess

Electrical & Electronics Engineering Graduate

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github.com/burjnx burjnx.github.io/personal-portfolio

PROFILE

Electrical and Electronics Engineering graduate from the University of Lagos who builds across hardware and software, from microcontrollers and solar-powered devices to machine learning and web apps. Hands-on, quick to learn, and keen to grow in a practical engineering or technology role.

PROFESSIONAL EXPERIENCE

Hardware and Embedded Systems Intern (SIWES) Apr 2025 – Sep 2025
Nithub (NITDA IT Hub), University of Lagos Lagos, Nigeria

- Trained in programming Arduino Uno and ESP32 microcontrollers in C++.
- Assembled and soldered a 6-relay ESP32 smart home board, and wrote the C++ code that switches each relay.
- Built and tested circuits with sensors and displays, including a motion sensor with an LCD readout.
- Learned KiCad for circuit schematics and PCB design, and how to read component datasheets.
- On the frontend team for Tasker, the hub's task-management web app, built the dashboard, projects, project detail and teams screens in React.js, working with Git branches and pull requests (live demo).

Media and Audio-Visual Volunteer Nov 2025
Nithub, University of Lagos Lagos, Nigeria

- Ran presentation laptops and projection for the three-day MIRG-ICAIR 2025 International Conference on AI and Robotics.

PROJECTS

ML-Based Power Management for a Solar-Powered IoT Node 2025 – 2026
Final-year project · Python, TensorFlow, TensorFlow Lite Micro, ESP32 (C++) View project

- A solar-powered sensor device that predicts its own battery level with machine learning and adjusts how often it takes readings, so it doesn't run out of power.
- Trained the model on a year of Lagos solar data; in simulation the battery never ran flat.
- Ran the model on an ESP32 with real temperature, humidity and light sensors.

Personal Portfolio Website 2025 – 2026
HTML, CSS, JavaScript burjnx.github.io/personal-portfolio

- Responsive site built from scratch with animated timelines, project pop-ups and photo galleries.

SKILLS

Engineering and Hardware
ESP32 and Arduino, circuit assembly and soldering, multimeter and lab test equipment, solar charging circuits, KiCad and Proteus (basics)

Programming and Web
Python, C/C++ (Arduino), JavaScript, HTML, CSS, React, Git and GitHub

Data and Machine Learning
TensorFlow/Keras, TensorFlow Lite Micro, NumPy, pandas, Google Colab

Office and Engineering Software
Microsoft Word, Excel and PowerPoint, MATLAB (coursework), Arduino IDE

EDUCATION

Bachelor of Science – Electrical and Electronics Engineering 2020 – 2026
University of Lagos Lagos, Nigeria

- Final examinations completed; results pending. Awaiting NYSC mobilisation.
- Relevant courses: Digital Signal Processing, Control Systems, Microprocessors, Computer Programming, Power Systems.